

HALEEMA TALLAT

Senior AI Engineer — LLM Systems, RAG & Agents

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Senior AI engineer designing production GenAI systems: RAG, agents, LLM inference, evaluation and governance.

EXPERIENCE

Senior AI Engineer, Agents & LLM Infrastructure — IgniteTech (via Crossover) Jul 2025 – Present

- Designed and built the platform's first production-grade social engine in ~3 months: a multi-tenant system with tenant isolation handling 1,000+ live customer interactions a day across 10–15 social and messaging channels, with a case processor, lakehouse, orchestrator, API and web layers covering ingestion, parallel processing, legacy-platform compatibility and publishing in each outlet's format. Scoped with product leads; its architecture playbook and runbooks became the template for 20 more engines.
- Scaled the channel engines on Kafka to 27 engines with KEDA autoscaling, Prometheus monitoring and graceful draining; re-architected Java monoliths into stateless TypeScript services, cutting memory 6–10× and startup ~20×.
- Architected a distributed platform running up to 500 concurrent coding agents: LangGraph on Ray, Pydantic-typed state and tools, Kafka events, Redis and PostgreSQL state, on Kubernetes on AWS.
- Established evaluation: Langfuse traces of every model and tool call, cost per run and tool-call errors in ClickHouse, and drafted replies scored by agent acceptance and edit rates.
- Built the LLM inference layer for agent runs and AI-drafted support replies, pairing self-hosted vLLM and Hugging Face models with the Anthropic API; cut cost and latency with batching, quantization and context management that makes agents reuse existing code.
- Designed the organisation's governed MCP architecture for Claude Code, Cursor and the agent platform, with OAuth2, capability controls, output validation and human approval for high-impact actions; onboarded teams, mentored engineers, and shipped on it an agent that checks live requests against schema-encoded rules.

Computer Vision Engineer — Brick & Mortar AI Mar – Oct 2025

- Showed a fine-tuned ResNet-12 few-shot classifier scored 89.7%, not the reported 46.7%, by tracing split leakage and protocol mismatches in its eval harness and rebuilding the held-out split.
- Cut cross-camera identity switches 20% and raised throughput 10–15× with segmentation-masked ReID (YOLOv8/v11) served on Triton via ONNX and TensorRT.

Computer Vision Engineer — NextBridge Nov 2024 – Mar 2025

- Built the end-to-end WebRTC control stack for a smart wheelchair: signalling server (SDP/ICE), joystick sender and on-chair Jetson ROS 2 receiver.
- Made the link fail safe with heartbeats, backoff reconnection and a watchdog that stops the chair on link loss; instrumented round-trip latency.

Machine Learning Team Lead — Powersoft19 Nov 2023 – Nov 2024

- Reduced field engineers' problem-resolution time 90% by architecting a production hybrid RAG system over technical documents across multiple data centres: Qdrant vector plus keyword search, tuned chunking and reranking, custom-trained domain adapters, measured with recall@k.
- Cut fault-detection turnaround from ~48 hours to ~30 minutes by leading a team of 3 engineers to replace manual visual inspection with automated anomaly detection on device telemetry, from data to production.
- Owned the data pipeline (Airflow ETL, PySpark over 1TB+ telemetry) on AWS and Google Cloud; optimised CUDA training with mixed precision.

Research Engineer — KICS, Center for Language Engineering, UET Nov 2022 – Nov 2023

- Reached 93% accuracy on NLP classification of enterprise text; reproduced RAG methods from the literature.
- Built Tajziyakar, a containerised OCR pipeline that extracts and flags TV news tickers at archive scale.

EDUCATION & AWARDS

BSc Electrical Engineering, UET Lahore, 2023 · CureMD Master Inventor Award, 2022 · COHACK hackathon winner, 2022

SKILLS

GenAI & RAG: RAG, hybrid search, reranking, embeddings, Qdrant, prompt engineering, structured outputs, fine-tuning and domain adaptation, vLLM, Hugging Face, Anthropic API

Agents & evaluation: LangGraph, LangChain, Ray, MCP, Pydantic, Langfuse, ClickHouse, Prometheus, MLflow

Cloud & data: AWS, Google Cloud, Docker, Kubernetes, Kafka, KEDA, Airflow, PySpark, PostgreSQL, GitHub Actions

ML & languages: Python, TypeScript, SQL, C++, PyTorch, CUDA, ONNX, TensorRT, Triton